

CSCI212 Lab — Quiz #3

1. In Java, You will develop **3** distinct classes, *Mob*, *Zombie*, *Creeper*, such that **Mob** is a super-class (or parent) of **Zombie** and **Creeper**.

(a) What **types** of classes are **Zombie** and **Creeper** with respect to **Mob**?

(b) In developing the **Mob** class you must include:

- Two **protected** attributes, *name* (*String*) and *health* (*int*)
- A **constructor** that takes in **2 arguments** *name* and *health* (of the same data-types in the previous step) and assigns these arguments to the respective attributes of the class.
- A **public** method titled *attack* that prints a single line message to the console:

```
1 // Example usage of attack();
2 Mob mob1 = new Mob("Enderman", 40);
3 mob1.attack();
4 // Prints: "Enderman attacks!"
5
6 Mob mob2 = new Mob("Iron Golem", 100);
7 mob2.attack();
8 // Prints: "Iron Golem attacks!"
```

- A **public** method titled *takeDamage* that takes in **1** argument *damage* (*int*), **subtracts** damage from health (the class attribute), and prints a single line message to the console indicating how much health the mob has after the update:

```
1 // Example usage of takeDamage();
2 mob1.takeDamage(20);
3 // Prints: "Enderman now has 20 HP";
4
5 mob2.takeDamage(30);
6 // Prints: "Iron Golem now has 70 HP";
```

(c) In developing the **Zombie** class you must include the following:

- **Zombie** *extends* **Mob**
- A **constructor** that takes in 2 arguments *name* (String) and *health* (int) and propagates them to the *super-class* using **super()**.
- A **public** method titled **attack** that *overrides* the method of the same name in the super-class (make sure to use the override annotation):

```
1 //Example usage of attack() for Zombie
2 Zombie zombie1 = new Zombie("Zombie Villager", 20);
3 zombie1.attack();
4 //Prints: "Zombie Villager swings its arms!"
5 Zombie zombie2 = new Zombie("Chicken Jockey", 20);
6 zombie2.attack();
7 //Prints: "Chicken Jockey swings its arms!"
```

(d) In developing the **Creeper** class you must include the following:

- **Creeper** *extends* **Mob**
- A **constructor** that takes in 2 arguments *name* (String) and *health* (int) and propagates them to the *super-class* using **super()**.
- A **public** method titled **attack** that *overrides* the method of the same name in the super-class (make sure to use the override annotation):

```
1 //Example usage of attack() for Creeper
2 Creeper creeper1 = new Creeper("Charged Creeper", 20);
3 creeper1.attack();
4 //Prints: "Charged Creeper explodes!"
5 Creeper creeper2 = new Creeper("Mutant Creeper", 10000);
6 creeper2.attack();
7 //Prints "Mutant Creeper explodes!"
```